

COUNTING HALLUCINATIONS IN DIFFUSION MODELS

Anonymous authors
Anonymous submission

ABSTRACT

Diffusion probabilistic models (DPMs) now generate high-fidelity images and videos, but visual realism does not guarantee factual consistency: a sample may appear plausible while violating structural regularities present in the data, such as a duplicated object instance or a hand with an impossible number of fingers. This paper provides a quantitative framework for a concrete and diagnostically useful class of such failures, which we term **counting hallucination**: generation of an inadmissible number of object instances or structured components when the reference data satisfy explicit count constraints. We introduce **CountHalluSet**, a dataset suite spanning ToyShape, SimObject, and RealHand, and instantiate a standardized evaluation protocol that converts count-level factuality into a measurable sample-level property. The protocol enables controlled analysis of how solver type, ODE solver order, sampling steps, and initial noise influence counting hallucination rates. Across these conditions, the results demonstrate that conventional sampling refinements improve count fidelity on simpler synthetic data but do not reliably enforce count constraints for structurally complex objects, with human hands providing the clearest case. We further show that widely used perceptual metrics, including Fréchet Inception Distance (FID), do not consistently track count errors, whereas non-counting failures align much more strongly with such metrics. Together, CountHalluSet, the accompanying protocol, and the empirical analysis establish a reproducible basis for diagnosing count-level factual failures in diffusion models and for developing generation methods that better preserve structured counts.

1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), or diffusion models, have become a dominant framework for generative modeling, with strong performance in text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). Their perceptual quality, however, does not eliminate a more specific form of factual error: generated samples can satisfy local visual expectations while contradicting structural regularities that are explicit in the data distribution. Canonical examples include morphologically intricate objects with invalid component counts, such as human hands with extra fingers (Narasimhaswamy et al., 2024) or animals with additional limbs (Borji, 2023).

Recent studies have substantially sharpened the qualitative understanding of diffusion-model failures, particularly in text-guided generation. For instance, Borji (2023) catalogued recurring deficiencies in text-to-image models, Liu et al. (2024) proposed adversarial search for undesirable generation behaviors, and Wu et al. (2023) introduced an iterative procedure for discovering and mitigating spurious correlations. These contributions identify important failure modes, but they do not provide a controlled quantitative protocol in which hallucination is defined by factual constraints of the reference distribution and measured consistently across sampling conditions.

Hallucination in diffusion models is therefore not merely a visual-quality problem; it is an evaluation problem centered on factual consistency (Aithal et al., 2024). Prompt-alignment errors can often be approximated by conditional agreement metrics such as CLIP score (Hessel et al., 2021) or VQA score (Huang et al., 2025; Zarei et al., 2024), and severe blur, distortion, or artifact formation can be treated as perceptual degradation. By contrast, hallucinated samples may remain counting-ready

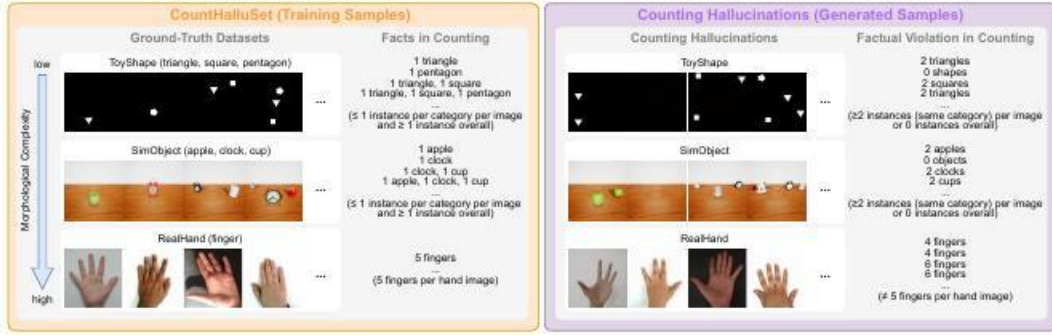


Figure 1: Training examples from **CountHalluSet** (left) and model-generated **counting hallucinations** under the corresponding dataset-specific count rules (right). ToyShape and SimObject permit at most one instance per category and require at least one instance overall, whereas RealHand requires exactly five fingers. A generated image is counted as a hallucination when it violates the applicable rule, such as by producing two apples or no object in SimObject, even though such configurations are absent from the reference dataset.

and visually plausible while violating the count, geometry, or physical structure that the dataset itself enforces. Because these discrepancies are fine-grained, current AI systems can miss them; even state-of-the-art large vision-language models such as GPT-4o struggle with the apparently simple task of counting fingers in a clear hand image. A standardized measurement protocol is thus necessary for converting qualitative observations into evidence that can guide model design, especially as generative models are increasingly expected to function as world models whose samples respect factual constraints (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

Among hallucinated attributes—including counts, colors, spatial relations, geometry, and physical realism—count errors are especially amenable to rigorous evaluation because the relevant factual constraint can be specified directly and checked at the sample level. We define **counting hallucination** as a generated sample whose object counts fall outside the values permitted by the training dataset. *In this setting, the number of objects in training data is explicitly predetermined. Consequently, a generated image is deemed hallucinated whenever its object counts deviate from these predefined values.* For RealHand, for example, a generated hand with six fingers is a counting hallucination because the reference images contain five-finger hands, as illustrated in the last row of Fig. 1. To study this failure mode across increasing visual complexity, we construct CountHalluSet, comprising ToyShape (triangle, square, pentagon), SimObject (mug, apple, clock), and RealHand (finger). Figure 1 shows representative reference examples together with the corresponding hallucination patterns.

The proposed protocol operationalizes counting hallucinations through dataset-specific admissible-count criteria and counting models that estimate the relevant object counts in generated images. Using this protocol, we quantify how denoising conditions—sampling steps, initial noise, ODE solver order, and solver type—alter counting hallucination rates in diffusion models. We further measure the relationship between these rates and perceptual metrics, including Fréchet Inception Distance (FID) Heusel et al. (2017) and MUSIQ Ke et al. (2021), thereby directly testing whether standard image-quality metrics capture count-level factual violations.

The paper makes three coordinated contributions:

- We introduce CountHalluSet and a standardized evaluation protocol that make count-level hallucinations measurable under explicit dataset-level constraints.
- We show through controlled denoising studies that common numerical refinements in diffusion sampling, including increasing sampling steps from 25 to 100, reduce counting hallucinations on simpler synthetic datasets but do not reliably mitigate them for human-hand generation.
- We establish that counting hallucination rates are weakly and inconsistently associated with standard perceptual metrics, whereas non-counting failure rates are strongly associated with those metrics; this empirical separation motivates joint-diffusion models, which substantially reduce both CHR and NCFR on RealHand.

2 RELATED WORK

2.1 HALLUCINATIONS IN DIFFUSION MODELS

Motivated by extensive work on hallucination in large language models (LLMs) (Kadavath et al., 2022; Ji et al., 2023; Huang et al., 2023), recent research has begun to examine analogous failures in diffusion models. In particular, Aithal et al. (2024) proposed the qualitative hypothesis that hallucinated samples may arise from interpolation between distinct data modes. This account supplies a plausible explanatory mechanism, but it leaves open the quantitative question of how often such failures occur and how their prevalence changes under standard sampling choices.

Our work addresses this measurement gap by isolating a hallucination class whose violations can be counted, separated from severe non-counting failures, and compared under controlled denoising conditions. The resulting formulation turns hallucination analysis from a primarily qualitative taxonomy into a reproducible empirical evaluation problem that exposes the effects of solver type, solver order, sampling steps, and initialization.

2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

Visual Question Answering (VQA) score (Huang et al., 2025; Zarei et al., 2024) and CLIP score (Hessel et al., 2021) are widely used to evaluate text-guided image generation by measuring agreement between a prompt and a generated image. Metrics that depend on language-based models, however, are not designed to certify factual consistency in hallucinated images. Pretrained language models exhibit a statistical tendency to hallucinate particular fact types, which imposes a ceiling on detection accuracy when evaluation relies solely on such models (Kalai & Vempala, 2024); human annotators also continue to outperform LLMs in classifying hallucinated textual content (Lin et al., 2022; Li et al., 2023).

Image-generation studies additionally rely on distributional and perceptual metrics, including FID (Heusel et al., 2017), IS (Salimans et al., 2016), Precision & Recall (Sajjadi et al., 2018), and MUSIQ (Ke et al., 2021). These metrics summarize image quality or distributional similarity through feature embeddings, often from a pretrained Inception-v3 model (Szegedy et al., 2016), but they do not explicitly encode count-level constraints. Accordingly, *whether these metrics can capture the severity of hallucinations in generated datasets* is an empirical question rather than a methodological guarantee. We answer this question for count-level factuality by measuring their relationship to counting hallucination rates under the CountHalluSet protocol.

3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

3.1 FORWARD (DIFFUSION) PROCESS

Forward Process. Given observed data $\mathbf{x}_0 \sim q_0(\mathbf{x}_0)$, Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) define a *forward process* (a.k.a. *diffusion process*) that transports data toward progressively noisier states by applying Gaussian perturbations over T discrete timesteps. At timestep t , the marginal distribution of the noisy variable $\mathbf{x}_t$ conditional on the original sample $\mathbf{x}_0$ is specified as:

$$q_{0:t}(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t | \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

where $t \in \{0, 1, \dots, T\}$ and $\{\beta_i\}_{i=1}^T$ denotes a predefined timestep-dependent noise schedule, commonly instantiated as a linear (Ho et al., 2020) or cosine (Nichol & Dhariwal, 2021) schedule. The coefficient $\bar{\alpha}_t$ determines the signal-to-noise ratio at step t : $\bar{\alpha}_0 = 1$ recovers the unperturbed sample, whereas $\bar{\alpha}_T \approx 0$ yields an almost pure-noise state. Equivalently, samples from Eq. (1) can be generated by writing:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + (1 - \bar{\alpha}_t)\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

For sufficiently large T , the terminal marginal $q_{0:T}(\mathbf{x}_T|\mathbf{x}_0)$ approaches the standard Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. Common DPM implementations (Ho et al., 2020; Nichol & Dhariwal, 2021) typically set T to 1000.

3.2 REVERSE (DENOISING) PROCESS.

Training Objective. The *reverse process* (a.k.a., *denoising process*) reconstructs data samples from the terminal noisy state $\mathbf{x}_T$ by sequentially removing noise. DPMs parameterize these reverse dynamics with a neural network $\epsilon_\theta(\mathbf{x}_t, t)$, the *noise prediction model*, which estimates the injected noise ϵ from the noisy state $\mathbf{x}_t$ and timestep t . The parameter vector θ is optimized with the standard denoising objective (Ho et al., 2020; Song et al., 2022):

$$\min_{\theta} \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\pi(t) \|\epsilon_\theta(\mathbf{x}_t, t) - \epsilon\|_2^2], \quad (3)$$

where $\pi(t) > 0$ denotes a weighting function.

Ancestral Sampling. Given a terminal noisy state $\mathbf{x}_T$, samples from the data distribution $q_0(\mathbf{x}_0)$ are generated by reversing the diffusion process across the corresponding timesteps; this stochastic procedure is commonly referred to as *ancestral sampling*:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \frac{1}{\sqrt{1-\beta_t}}(\mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)), \beta_t \mathbf{I}), \quad (4)$$

where $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ is the unknown *score function* (Song et al., 2022).

In practice, DPMs replace the scaled score function $-\sqrt{(1-\bar{\alpha}_t)} \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ with the learned noise prediction model. This substitution produces the following ancestral update, which is equivalent to the denoising parameterization in Eq. (3):

$$\mathbf{x}_{t-1} = \mu_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon, \quad \mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1-\beta_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t). \quad (5)$$

ODE-Based Accelerated Sampling. The ancestral sampler in Eq. (4) is stochastic (Kloeden et al., 1992). A deterministic counterpart is obtained through the ordinary differential equation (ODE) formulation known as the *probability flow ODE* (Song et al., 2022):

$$\frac{d\mathbf{x}(\tau)}{d\tau} = f(\tau)\mathbf{x}(\tau) + h(\mathbf{x}(\tau), \tau), \quad f(\tau) = -\frac{1}{2}\beta(\tau), \quad h(\mathbf{x}(\tau), \tau) = -\frac{1}{2}\beta(\tau) \nabla_{\mathbf{x}(\tau)} \log q_\tau(\mathbf{x}(\tau)), \quad (6)$$

where $\tau \in [T, 0]$. Let $\mathbf{x}(\tau)$ denote the ideal trajectory satisfying this ODE under the true score function $h(\mathbf{x}(\tau), \tau)$ and initialized from ideal diffused data $\mathbf{x}(T)$ drawn from $q_T(\mathbf{x}(T))$. We discretize time as $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, with terminal index $\tau_N = 0$. The variation-of-constants formula yields the exact one-step solution:

$$\mathbf{x}(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h(\mathbf{x}(u), u) du \quad (7)$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state-transition operator for the linear dynamics $\frac{dz}{d\tau} = f(\tau)z$. The step size $\Delta t > 0$ determines the number of sampling steps, $T/\Delta t$, and permits ODE-based samplers to accelerate inference by using larger integration steps than ancestral sampling. Because $\mathcal{G}(t_2, t_1)$ has a closed form, accelerated ODE samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022; 2023) concentrate approximation error in the nonlinear integral involving h , with accuracy governed by solver order and the number of sampling steps. Appendix A.1 to A.3 provides the corresponding theoretical analysis.

4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

This section defines the evaluation framework used to quantify counting hallucinations. We first specify dataset-level counting criteria and the corresponding violation event, then instantiate the definition in the CountHalluSet suite. For each dataset, we state the admissible counts, identify the hallucination cases induced by those criteria, and describe the standardized measurement protocol used in the experiments.

4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

Counting criteria are explicit constraints on object counts in the reference data; counting hallucinations are generated samples that violate those constraints while remaining visually suitable for counting.

Counting criteria in a training dataset. Let $\mathcal{D}_{\text{ref}}$ denote a training dataset. For any reference sample $\mathbf{x} \in \mathcal{D}_{\text{ref}}$, the count of objects from category c , denoted $N_c(\mathbf{x})$, must lie in a predefined set $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$ (e.g., $\{0, 1\}$ or $\{1, 2, 3\}$), i.e., $N_c(\mathbf{x}) \in \mathcal{S}_c$. Each valid reference sample is also required to contain at least one object, i.e. $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$, where C is the set of categories under evaluation.

Counting hallucinations as deviations from the counting criteria. Given a generated sample $\hat{\mathbf{x}}$, we say that $\hat{\mathbf{x}}$ exhibits a *counting hallucination* with respect to $\mathcal{D}_{\text{ref}}$ precisely when both of the following conditions hold:

1. **Sufficient visual quality for counting.** The image $\hat{\mathbf{x}}$ must have sufficient visual quality for the background and any present objects to be clearly discernible. This requirement distinguishes count-level factual violations from severe degradations that make counting ill-posed. We define a binary variable $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \in \{0, 1\}$, the *counting-ready indicator*, where 1 denotes that the image is suitable for counting and 0 denotes that it is not. In practice, this indicator may be obtained from human annotation, from a classifier calibrated to human annotations, or from other automated criteria.
2. **Violation of counting facts.** A counting-fact violation occurs when there exists a category c such that $N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$, or when the image contains no objects (i.e., $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$) despite the dataset-level requirement that each sample contain at least one object. A counting model supplies the empirical evidence for these violations by predicting object counts in generated images.

Thus, a counting hallucination (CH) is identified precisely when the sample is counting-ready ($\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) = 1$) and its predicted counts violate the counting criteria ($\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$ or $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$). Formally,

$$\mathbb{I}_{\text{CH}}(\hat{\mathbf{x}}) = \mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \wedge \left(\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c \vee \sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0 \right).$$

4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

The CountHalluSet suite consists of three datasets designed to evaluate counting hallucinations across increasing visual complexity: ToyShape, SimObject, and RealHand. Examples from each dataset and corresponding counting hallucinations are shown in Fig. A1 and A2. Additional statistical summaries are provided in Appendix A.4.

4.2.1 TOYSHAPE

Dataset description. The ToyShape dataset comprises 30,000 images containing three geometric shape categories: *triangle*, *square*, and *pentagon*. All shapes are white, have equal area of 120 pixels, and do not overlap within an image. The counting criteria require each image to contain at most one instance of each shape category and at least one shape in total, as shown in the top-left of Fig. 1. Formally, we have $C = \{\text{triangle}, \text{square}, \text{pentagon}\}$. For each class $c \in C$, we set $N_c(\mathbf{x}) \in \mathcal{S}_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Counting hallucinations. Under these criteria, any image generated by a model trained on ToyShape that contains two or more shapes from the same category is a counting hallucination, such as the image with two pentagons in the top-right region of Fig. 1. Empty images containing only the black background and no shapes are also classified as counting hallucinations. Examples are provided in Appendix A.5.

Counting model used in ToyShape. We use a counting model to identify count errors directly, without a counting-ready indicator, because the simplicity of ToyShape makes severe degradations rare. Specifically, we fine-tune a ResNet-50 (He et al., 2016) on more than 400k ToyShape samples in which each category appears 0–3 times per image (i.e., $\mathcal{S}_c = \{0, 1, 2, 3\}$), so that the classifier

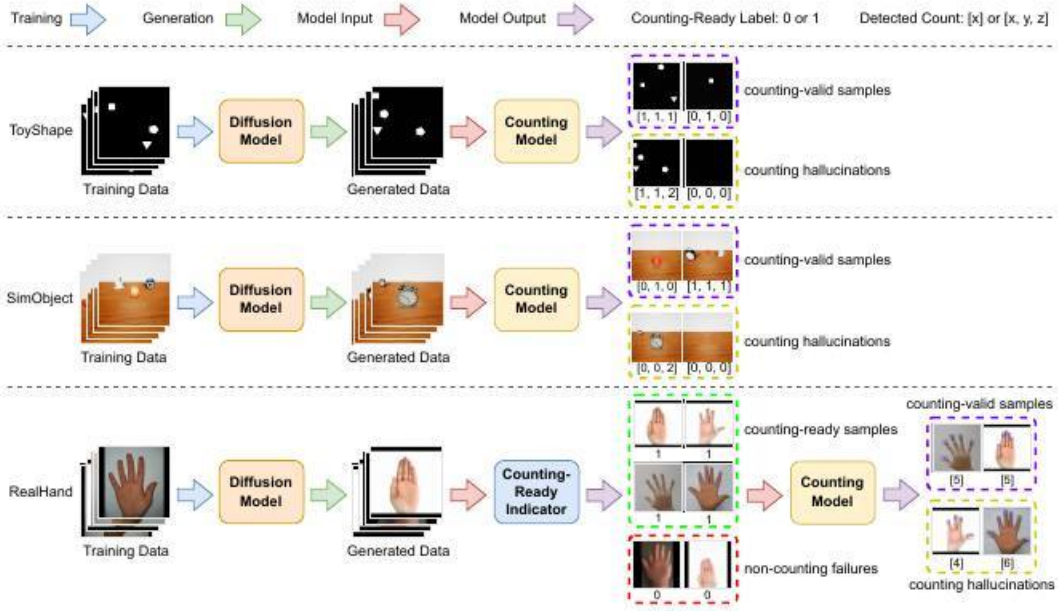


Figure 2: Evaluation procedure for the CountHalluSet suite. For ToyShape and SimObject, generated images are evaluated directly by the corresponding counting models. For RealHand, a counting-ready indicator (a binary classifier aligned with human judgments) first separates countable images from non-counting failures, such as severely deformed fingers. Count labels are ToyShape: *triangle*, *square*, *pentagon*; SimObject: *mug*, *apple*, *clock*; RealHand: *finger*.

covers both counting-valid samples and counting hallucinations. The resulting model achieves over 99.9% counting accuracy on generated data.

Quantification of counting hallucinations. For a diffusion model trained on ToyShape, we generate 30,000 samples, matching the size of the training dataset. The counting model then predicts the number of objects in each generated image. Images whose predicted counts violate the ToyShape criteria are identified as counting hallucinations, as illustrated in Fig. 2.

4.2.2 SIMOBJECT

Dataset description. The SimObject dataset contains 30,000 rendered images of everyday objects. It includes three object categories, *mug*, *apple*, and *clock*, each with 10 distinct intra-class variations. Objects are randomly placed on a wooden tabletop under fixed lighting conditions, with minimal mutual occlusion when occlusion occurs, as shown in the middle-left panel of Fig. 1. The dataset is synthetically generated in Unreal Engine 5 (Epic Games, 2024), using photorealistic rendering with simulation of object geometry, material attributes, lighting, and reflections. Its counting criteria mirror those of ToyShape (see Sec. 4.2.1). Formally, we have $C = \{mug, apple, clock\}$. For each class $c \in C$, we set $N_c(x) \in S_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(x) \geq 1$.

Quantification of counting hallucinations. Because SimObject and ToyShape share the same admissible-count structure, the definition of counting hallucinations, the construction of the counting model, and the quantification protocol follow the ToyShape procedure, as shown in Fig. 2.

4.2.3 REALHAND

Dataset description. To evaluate counting hallucinations on real visual structure, we construct RealHand, a dataset of 5,050 human hand images sourced from the 11k Hands dataset (Afifi, 2019), Kaggle¹, and Roboflow². We aggregate raw images from these sources and remove images with blur or artifacts. Each retained image depicts a human hand with five fingers, including both dorsal (back) and palmar (front) views under varied lighting conditions and backgrounds (e.g., white surfaces,

¹<https://www.kaggle.com/>

²<https://universe.roboflow.com/>

blackboards, walls), as shown in the lower-left of Fig. 1. The counting criteria require exactly five fingers, i.e., $C = \{\text{finger}\}$ with $N_c(x) \in S_c = 5$ for each $c \in C$, and $\sum_{c \in C} N_c(x) = 5$.

Counting hallucinations and non-counting failures. Realistic hand generation produces failures beyond count errors, including blurriness, severely deformed fingers, and visual artifacts. We therefore use a counting-ready indicator (i.e., a binary classifier) to separate images that can be meaningfully counted from these non-counting failures. Examples of counting hallucinations and non-counting failure samples are provided in Appendix A.7.

Counting-ready indicator and counting model used in RealHand. To separate counting hallucinations from these other failure modes, we finetune a MaxViT (Tu et al., 2022) model on 2.5k annotated generated images and use it as a counting-ready indicator aligned with human judgments. For fingertip counting, we finetune a YOLO-12 (Tian et al., 2025) model on 2k fingertip-annotated generated images. Based on human evaluation of 500 randomly sampled images by three annotators, the indicator and counting model achieve over 96% and 99% accuracy, respectively. Examples of YOLO detection results are shown in Appendix A.6.

Quantification of counting hallucinations. Unlike ToyShape and SimObject, where all generated images can be evaluated directly because severe failures are rare, RealHand requires an additional filtering stage. We first apply the counting-ready indicator to select hand images with sufficient visual quality, and then apply the counting model to predict the number of fingers, as shown in Figure 2.

5 EXPERIMENTS

5.1 IMPLEMENTATION DETAILS

Training of Diffusion Models. For ToyShape, we train a DDPM (Ho et al., 2020) from scratch; for SimObject and RealHand, we fine-tune latent diffusion models (LDMs) (Rombach et al., 2022) pretrained on CelebA-HQ (Karras et al., 2018). The diffusion horizon is $T = 1,000$ with a standard linear noise schedule. Images are resized to 128×128 for DDPMs and 256×256 for LDMs. Models are trained with Adam (Kingma, 2015) using a learning rate of 0.0001 for 150k, 300k, and 80k steps on ToyShape, SimObject, and RealHand, respectively, with an effective batch size of 256.

Inference of Diffusion Models. For ancestral sampling (i.e., DDPM), we use 1,000 sampling steps. For ODE-based sampling, we evaluate both first- and second-order methods: DPM-Solver-1 (equivalent to DDIM (Song et al., 2021)) as the first-order solver and DPM-Solver-2 (Lu et al., 2022) as the second-order solver. Each ODE solver is evaluated at three commonly used denoising budgets: 25, 50, and 100 steps. For RealHand, the counting model uses a YOLO detector with confidence threshold 0.3 and IoU threshold 0.1. Unless otherwise noted, each evaluation generates the same number of samples as the corresponding training set, and all reported results are averaged over three random seeds.

5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

Table 1 reports counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) under controlled denoising conditions across ToyShape, SimObject, and RealHand. The comparisons isolate the effects of solver type, ODE solver order, sampling-step budget, and initial-noise distribution. Five empirical patterns are central to the paper’s argument.

(1) **Increasing sampling steps reduces counting hallucinations in synthetic datasets but exacerbates them in RealHand.** As shown in Table 1, under ODE solver settings of 25, 50, and 100 steps, increasing the number of sampling steps generally lowers CHR on ToyShape and SimObject but tends to raise CHR on RealHand, with the exception of DPM-Solver-2 at 100 steps. This divergence indicates that finer solver granularity benefits simpler, more structured synthetic distributions, whereas RealHand exposes a more complex regime in which additional denoising steps can amplify count-level inconsistencies rather than remove them.

(2) **Employing higher-order ODE solvers effectively mitigates non-counting failures but exacerbates counting hallucinations in RealHand under the standard inference steps.** On RealHand, the comparison between DPM-Solver-1 and DPM-Solver-2 at standard sampling budgets such as

Table 1: Counting hallucination rate (CHR; counting hallucinations divided by total generated samples) under solver configurations and initial-noise settings on ToyShape, SimObject, and RealHand. For RealHand, non-counting failure rate (NCFR) reports failures identified by the counting-ready indicator, and total failure rate (TFR) is the sum of CHR and NCFR. *Diffused* denotes ground-truth initial noise, whereas *Normal* denotes standard Gaussian noise.

Solver Name	Sampling Steps	CHR (ToyShape)		CHR (SimObject)		CHR (RealHand)		NCFR (RealHand)		TFR (RealHand)	
		<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>
DPM-Solver-1 (Song et al., 2021)	25	2.43	3.47	9.27	9.95	12.71	12.95	16.18	18.06	28.90	31.01
	50	1.83	2.79	8.53	9.27	13.82	13.85	11.32	12.51	25.14	26.36
	100	1.56	2.32	8.04	8.90	14.29	14.55	9.54	10.63	23.83	25.19
DPM-Solver-2 (Lu et al., 2022)	25	2.45	3.81	8.16	8.19	14.23	14.48	8.37	9.33	22.60	23.82
	50	1.76	2.86	7.90	7.95	16.15	15.99	6.38	7.22	22.53	23.21
	100	1.41	2.24	7.89	7.89	15.06	15.43	7.82	8.94	22.88	24.38
DDPM (Ho et al., 2020)	1000	0.63	0.64	4.98	4.99	10.82	10.75	2.07	2.39	12.88	13.14

25 and 50 steps reveals a systematic trade-off: the higher-order solver substantially reduces NCFR while simultaneously increasing CHR. A related pattern appears when the sampling budget increases for both DPM-Solver-1 and DPM-Solver-2. Thus, additional computation and numerically stronger sampling conditions improve perceptual viability, as reflected in lower NCFR, but they do not by themselves enforce factual correctness of finger counts.

(3) **DDPM substantially outperforms ODE solvers in mitigating both counting hallucinations and non-counting failures.** Across all comparisons in Table 1, ancestral sampling (i.e., DDPM) achieves the lowest CHR, NCFR, and TFR. Within the evaluated settings, DDPM therefore provides the strongest empirical reference point for low failure rates and demonstrates that stochastic ancestral sampling can be especially effective for minimizing both count-level and non-counting failures.

(4) **A better initial noise configuration can reduce both counting hallucinations and non-counting failures.** As shown in Table 1, using “diffused” noise (i.e., ground-truth initial noise) generally yields lower CHR, NCFR, and TFR than using “normal” noise (i.e., standard Gaussian noise). This pattern is consistent with the interpretation that diffused noise better matches the initial-noise distribution encountered during training, so a more informed initialization can reduce both counting hallucinations and non-counting failures.

(5) **Greater morphological complexity in countable objects increases the likelihood of counting hallucinations.** Comparing CHR across ToyShape, SimObject, and RealHand shows that count errors become more frequent as the morphology of the countable target becomes more complex. Fine-grained components and spatial dependencies in complex objects therefore pose a stronger challenge for diffusion models, increasing the likelihood of additions or omissions in generated samples.

5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

Counting hallucination is a form of factualness hallucination, so it is natural to ask: *Does a better perceptual metric, such as a lower FID, necessarily imply fewer counting hallucinations by reflecting a smaller distribution gap between generated and training data?* The answer is **not necessarily**.

To examine this relationship, we perform correlation analysis between CHR and perceptual metrics, as well as between the NCFR and the same metrics. We focus on two widely used perceptual measures: a distribution-based metric, FID (Heusel et al., 2017), and an instance-based metric, MUSIQ (Ke et al., 2021). For correlation estimation, we adopt three complementary measures: Pearson correlation (Lee Rodgers & Nicewander, 1988) (linear dependence), Spearman rank correlation (Spearman, 1904) (monotonic dependence), and Distance Correlation (Székely et al., 2007) (general nonlinear dependence). The following evidence supports our conclusion.

✦ **The correlations between CHR and perceptual metrics are dataset-dependent and solver-dependent rather than intrinsic.** As shown in the first and third rows of Table 2, the Pearson correlations between CHR and FID vary substantially across datasets (i.e., positive for SimObject, negative for RealHand, and positive again for PTI-Hand), while the corresponding Spearman correlations shift from positive to negative to negative. Distance correlations also decrease from 0.82

Table 2: Correlation of counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) with perceptual metrics, including FID and MUSIQ. Results for PTI-Hand are obtained by prompting a pretrained Stable-Diffusion-3.5-Medium model with ten hand-generation prompts to produce 5050 samples, matching the size of the RealHand dataset. FID for PTI-Hand is computed between RealHand images and the model-generated samples. Here, p denotes the p-value assessing the statistical significance of the correlation r or ρ . $\dagger$ means DDPM results integrated with those from various ODE solvers across sampling conditions.

Metric	CHR vs. FID						NCFR vs. FID						CHR vs. MUSIQ						NCFR vs. MUSIQ					
	SimObject	RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	
Pearson (r)	0.87	0.01	-0.97	<0.01	0.20	0.69	0.99	<0.01	0.88	<0.01	0.93	<0.01	-0.02	0.95	-0.97	<0.01	-0.99	<0.01						
Pearson † (r)	0.49	0.14	-0.53	0.13	—	—	0.96	<0.01	—	—	0.46	0.20	—	—	-0.98	<0.01	—	—						
Spearman (ρ)	0.84	<0.01	-0.97	<0.01	-0.14	0.78	0.99	<0.01	0.92	<0.01	0.80	0.01	0.42	0.33	-0.88	<0.01	-0.46	0.29						
Spearman † (ρ)	0.75	0.01	-0.51	0.15	—	—	0.95	<0.01	—	—	0.40	0.28	—	—	-0.91	<0.01	—	—						
Distance Correlation (r_d)	0.89	—	0.97	—	0.47	—	0.99	—	0.91	—	0.93	—	0.42	—	0.97	—	0.99	—						
Distance Correlation † (r_d)	0.82	—	0.74	—	—	—	0.95	—	—	—	0.71	—	—	—	0.97	—	—	—						

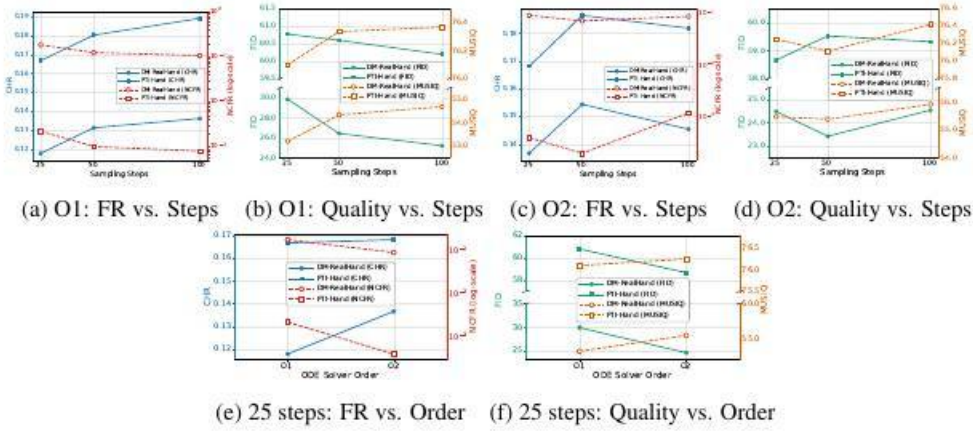


Figure 3: Comparison of trend consistency between DM-RealHand (ours) and PTI-Hand (pretrained Stable-Diffusion-3.5-Medium). O1 and O2 denote first- and second-order ODE solvers (DPM-solver-1 / DPM-solver-2 for DM-RealHand; Euler / Heun for PTI-Hand). FR metrics include CHR and NCFR; quality metrics include FID and MUSIQ. Across variations in sampling steps and solver order, both models exhibit highly consistent trends.

(SimObject) to 0.74 (RealHand) to 0.47 (PTI-Hand), suggesting that more complex generative content weakens the association between CHR and FID. Moreover, integrating DDPM results with those from ODE solvers further attenuates these correlations.

✎ **The correlations between NCFR and perceptual metrics are consistently strong and positive.** In contrast to the weak correlations observed between CHR and perceptual metrics, the correlations between NCFR and perceptual metrics is strong and robust across all settings. For example, all correlation coefficients are over 0.88 in NCFR vs. FID, with very small p-values, indicating high statistical significance.

5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

In this section, we evaluate counting hallucinations in a large-scale pretrained text-to-image model. Specifically, we assess Stable-Diffusion-3.5-Medium (Esser et al., 2024) using the counting-ready indicator and counting model introduced in Sec. 4.2.3. More details are provided in Appendix A.8. As shown in Figure 3, the way solver type and sampling steps affect counting hallucinations in large pretrained diffusion models aligns closely with the trends revealed by our controlled RealHand experiments. This consistency highlights the validity of the findings presented in Sec. 5.2 and underscores the broader applicability of our evaluation protocol. Correlation results in Table 2 further support this conclusion. In specific, weak correlations between CHR and perceptual metrics and strong correlations between NCFR and perceptual metrics are also observed. Together, these results

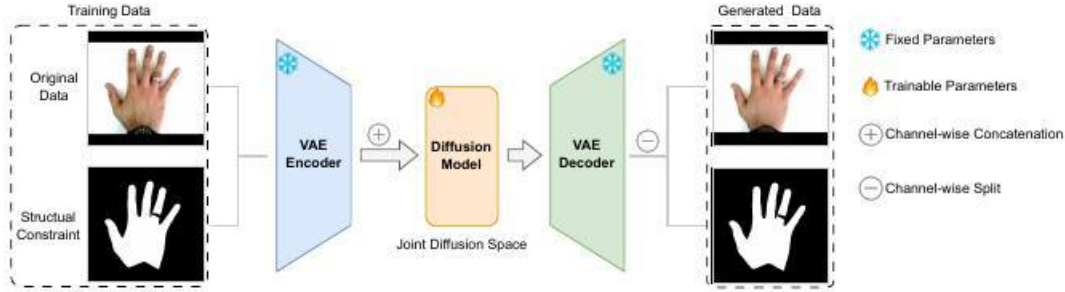


Figure 4: Diagram of the proposed joint-diffusion model (JDM). The model performs channel-wise concatenation of the latent representations of hand RGB images and their corresponding segmentation masks. This joint learning paradigm encourages anatomically plausible hand poses and promotes correct finger topology during generation, resulting in reduced CHR and NCFR.

Table 3: Comparison of failure rates between LDM and our proposed JDM on the RealHand dataset across different ODE solvers and sampling steps. Lower values indicate better performance. For JDM, although the model jointly generates hand RGB images and segmentation masks, FID is computed solely on the RGB outputs to ensure comparability with baselines.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fr�chet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
LDM	12.95	13.85	14.55	14.48	15.99	15.43	18.06	12.51	10.63	9.33	7.22	8.94	29.86	26.43	25.21	24.48	23.42	24.54
JDM (ours)	11.41	10.67	10.51	10.12	9.66	10.07	2.94	2.42	2.29	3.51	4.63	2.26	37.69	35.51	34.68	34.63	32.07	33.89

demonstrate the robustness and generality of our findings and show that the proposed evaluation protocol transfers reliably to pretrained text-to-image models.

5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

The lower counting hallucination rates observed for simpler countable objects (see Observation 5 in Sec. 5.2) suggest that diffusion models more reliably capture counting-based factuality when the underlying visual structure exhibits low morphological complexity (e.g., masked objects). This finding raises an important question: if diffusion models were to denoise general visual representations together with explicit, simple structural constraints that are easier to model, such as object masks that encode basic topology and counts, within a shared latent space, could such a joint learning paradigm mitigate counting hallucinations? We refer to such models that jointly learn both general visual representations and structural constraints during denoising as **joint-diffusion models (JDM)**.

Specifically, we perform a channel-wise concatenation of the latent representation of hand mask with that of the corresponding RGB image, as shown in Fig. 4. Then we finetune the UNet pretrained on CelebA-HQ (Karras et al., 2018) (following the RealHand configuration described in Sec. 5.1) using the RealHand images and their associated masks, extracted using SAM-2 (Ravi et al., 2024). To accommodate both modalities, the input and output channels of the denoising UNet are expanded to match the combined latent dimensionality of the image and mask latents (i.e., 3+3=6 channels).

As shown in Table 3, JDM consistently achieves lower failure rates than the vanilla LDM, including CHR and NCFR, across diverse sampling conditions, highlighting its effectiveness in accurately generating structurally intricate objects that are prone to counting ambiguities. On the other hand, JDM yields higher FID. This degradation likely stems from the increased denoising burden introduced by jointly modeling two modalities under a small-scale training regime, which forces the model to trade off perceptual quality for improved structural realism. We also observe promising reductions in counting hallucinations when applying JDM to large-scale pretrained diffusion models under carefully controlled finetuning settings. More details, results and discussion can be found in Appendix A.9.

6 CONCLUSION

In this work, we take the first step toward systematically quantifying hallucinations in diffusion models and offer new insights into improving factual generation in this field. Focusing on counting hallucinations, we introduce **CountHalluSet** and a practical and reproducible methodology for measuring counting hallucinations in practice. Through comprehensive and controlled quantification experiments, we show that commonly adopted numerical strategies often fail to mitigate counting hallucinations in generating human hands. Moreover, our correlation analyses demonstrate that widely used perceptual metrics (e.g., FID and MUSIQ) are unable to consistently reflect the severity of counting hallucinations. Experiments on large-scale pretrained text-to-image models further corroborate both the generality and the validity of our findings.

REFERENCES

- Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore, Andrew J Ballard, Joshua Bambrick, et al. Accurate structure prediction of biomolecular interactions with alphafold 3. *Nature*, pp. 493—500, 2024.
- Mahmoud Afifi. 11k hands: Gender recognition and biometric identification using a large dataset of hand images. *Multimedia Tools and Applications*, 78:20835–20854, 2019.
- Sumukh K Aithal, Pratyush Maini, Zachary C Lipton, and J Zico Kolter. Understanding hallucinations in diffusion models through mode interpolation. *arXiv preprint arXiv:2406.09358*, 2024.
- Ali Borji. Qualitative failures of image generation models and their application in detecting deepfakes. *Image and Vision Computing*, 137:104771, 2023.
- Jake Bruce, Michael D Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes, Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, et al. Genie: Generative interactive environments. In *ICML*, 2024.
- Thomas M Cover and Joy A Thomas. *Elements of Information Theory*. John Wiley & Sons, 2012.
- Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *NeurIPS*, 34:8780–8794, 2021.
- Jingtao Ding, Yunke Zhang, Yu Shang, Yuheng Zhang, Zefang Zong, Jie Feng, Yuan Yuan, Hongyuan Su, Nian Li, Nicholas Sukiennik, et al. Understanding world or predicting future? a comprehensive survey of world models. *ACM Computing Surveys*, 2024.
- Epic Games. Unreal engine, 2024. URL <https://www.unrealengine.com>.
- Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *ICML*, pp. 12606–12633, 2024.
- David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2(3), 2018.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pp. 770–778, 2016.
- Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. Clipscore: A reference-free evaluation metric for image captioning. In *EMNLP*, pp. 7514–7528, 2021.
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *NeurIPS*, 30, 2017.
- Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. In *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*, pp. 1–14, 2021.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *NeurIPS*, 33: 6840–6851, 2020.

- Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J Fleet. Video diffusion models. *NeurIPS*, 35:8633–8646, 2022.
- Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Multilayer feedforward networks are universal approximators. *Neural networks*, 2(5):359–366, 1989.
- Kaiyi Huang, Chengqi Duan, Kaiyue Sun, Enze Xie, Zhenguo Li, and Xihui Liu. T2i-compbench++: An enhanced and comprehensive benchmark for compositional text-to-image generation. *IEEE TPAMI*, 2025.
- Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, et al. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *arXiv preprint arXiv:2311.05232*, 2023.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM computing surveys*, 55(12):1–38, 2023.
- John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, et al. Highly accurate protein structure prediction with alphafold. *Nature*, 596(7873):583–589, 2021.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- Adam Tauman Kalai and Santosh S Vempala. Calibrated language models must hallucinate. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, pp. 160–171, 2024.
- Tero Karras, Timo Aila, Samuli Laine, and Jaakko Lehtinen. Progressive growing of gans for improved quality, stability, and variation. In *ICLR*, 2018.
- Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. Musiq: Multi-scale image quality transformer. In *ICCV*, pp. 5148–5157, 2021.
- Diederik P Kingma. Adam: A method for stochastic optimization. In *ICLR*, pp. 1–15, 2015.
- Peter E Kloeden, Eckhard Platen, Peter E Kloeden, and Eckhard Platen. *Stochastic differential equations*. Springer, 1992.
- Zhifeng Kong, Wei Ping, Jiaji Huang, Kexin Zhao, and Bryan Catanzaro. Diffwave: A versatile diffusion model for audio synthesis. In *ICLR*, pp. 1–17, 2021.
- Joseph Lee Rodgers and W Alan Nicewander. Thirteen ways to look at the correlation coefficient. *The American Statistician*, 42(1):59–66, 1988.
- Junyi Li, Xiaoxue Cheng, Wayne Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. Halueval: A large-scale hallucination evaluation benchmark for large language models. In *EMNLP*, pp. 6449–6464, 2023.
- Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. In *ACL*, pp. 3214–3252, 2022.
- Luping Liu, Yi Ren, Zhijie Lin, and Zhou Zhao. Pseudo numerical methods for diffusion models on manifolds. In *ICLR*, pp. 1–24, 2022.
- Qihao Liu, Adam Kortylewski, Yutong Bai, Song Bai, and Alan Yuille. Discovering failure modes of text-guided diffusion models via adversarial search. In *ICLR*, pp. 1–25, 2024.
- Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. Dpm-solver: A fast ode solver for diffusion probabilistic model sampling in around 10 steps. *NeurIPS*, 35:5775–5787, 2022.
- Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. Dpm-solver++: Fast solver for guided sampling of diffusion probabilistic models. In *ICLR*, pp. 1–19, 2023.

- Shitong Luo and Wei Hu. Diffusion probabilistic models for 3d point cloud generation. In *CVPR*, pp. 2837–2845, 2021.
- Supreeth Narasimhaswamy, Uttaran Bhattacharya, Xiang Chen, Ishita Dasgupta, Saayan Mitra, and Minh Hoai. Handdiffuser: Text-to-image generation with realistic hand appearances. In *CVPR*, pp. 2468–2479, 2024.
- Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *ICML*, pp. 8162–8171, 2021.
- Michael A Nielsen. *Neural networks and deep learning*, volume 25. Determination press San Francisco, CA, USA, 2015.
- Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *CVPR*, pp. 10684–10695, 2022.
- Mehdi SM Sajjadi, Olivier Bachem, Mario Lucic, Olivier Bousquet, and Sylvain Gelly. Assessing generative models via precision and recall. *NeurIPS*, 31, 2018.
- Tim Salimans, Ian Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, and Xi Chen. Improved techniques for training gans. *NeurIPS*, 29, 2016.
- Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *ICML*, pp. 2256–2265, 2015.
- Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *ICLR*, pp. 1–22, 2021.
- Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *ICLR*, pp. 1–36, 2022.
- C Spearman. The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1):72–101, 1904.
- Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *CVPR*, pp. 2818–2826, 2016.
- Gábor J Székely, Maria L Rizzo, and Nail K Bakirov. Measuring and testing dependence by correlation of distances. *The Annals of Statistics*, pp. 2769–2794, 2007.
- Yunjie Tian, Qixiang Ye, and David Doermann. Yolov12: Attention-centric real-time object detectors. *arXiv preprint arXiv:2502.12524*, 2025.
- Zhengzhong Tu, Hossein Talebi, Han Zhang, Feng Yang, Peyman Milanfar, Alan Bovik, and Yinxiao Li. Maxvit: Multi-axis vision transformer. In *ECCV*, pp. 459–479, 2022.
- Arash Vahdat, Francis Williams, Zan Gojcic, Or Litany, Sanja Fidler, Karsten Kreis, et al. Lion: Latent point diffusion models for 3d shape generation. *NeurIPS*, 35:10021–10039, 2022.
- Shirley Wu, Mert Yuksekgonul, Linjun Zhang, and James Zou. Discover and cure: Concept-aware mitigation of spurious correlation. In *ICML*, pp. 37765–37786, 2023.
- Dmitry Yarotsky. Universal approximations of invariant maps by neural networks. *Constructive Approximation*, 55(1):407–474, 2022.
- Arman Zarei, Keivan Rezaei, Samyadeep Basu, Mehrdad Saberi, Mazda Moayeri, Priyatham Katakinda, and Soheil Feizi. Understanding and mitigating compositional issues in text-to-image generative models. *arXiv preprint arXiv:2406.07844*, 2024.
- Qinsheng Zhang and Yongxin Chen. Fast sampling of diffusion models with exponential integrator. In *ICLR*, pp. 1–33, 2023.

A APPENDIX

A.1 POOF OF THE BOUND OF TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

For the joint distributions of the ideal reverse process $q(\mathbf{x}_{T:0})$ and the true reverse process $p_\theta(\mathbf{x}_{T:0})$, their KL divergence is defined as:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

For both reverse processes $q(\mathbf{x}_{T:0})$ and $p_\theta(\mathbf{x}_{T:0})$, we can decompose them as follows, leveraging the Markov property:

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

By applying the chain rule to decompose the KL divergence, we have:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[\mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[\log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

Then, according to the Data Processing Inequality (Cover & Thomas, 2012), marginalizing (or discarding some random variables) will not increase the KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

The bound of total accumulated distributional error can be obtained by combining Eq. (A.4) and Eq. (A.5):

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

where equality (=) would strictly hold only in the ideal case where $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$ and $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ for all t and $\mathbf{x}_t$, which means the diffusion model can accurately recover the underlying data distribution $q(\mathbf{x}_0)$. This upper bound for the overall KL divergence ($\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0))$) between true data and model-generated data is identical to the variational bound derived in DDPM (Ho et al., 2020) when data entropy $H(\mathbf{x}_0)$ is excluded, thereby suggesting that optimizing DDPM’s variational loss directly tightens this explicit upper bound on the discrepancy between trued data and model-generated data.

Diffused-prior gap. The first source of errors arises from the mismatch between the distribution of the ground-truth diffused data $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T|\mathbf{x}_0)$ and the prior distribution used for sampling $p_\theta(\mathbf{x}_T)$. This mismatch stems from the fact that the number of diffusion steps T is finite in practice. The discrepancy can be formally quantified using a general distributional metric:

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

where $\mathcal{D}_{\text{KL}}$ denotes the Kullback–Leibler (KL) divergence. Alternatively, other appropriate metrics such as FID or the Wasserstein distance can be employed to evaluate this distributional discrepancy. The diffused-prior gap, present from the outset, induces a deviation that persists throughout the denoising process, ultimately resulting in a sustained output error.

Noise model prediction error. The second source of errors stems from the noise prediction model. Because the model’s estimate $\epsilon_\theta(\mathbf{x}_t, t)$ only approximates the true noise ϵ , as guaranteed in principle by the *Universal Approximation Theorem* (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022). Let $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$ represent this prediction error. Based on Eq. (5), any discrepancy in ϵ_θ induces a corresponding deviation in the predicted mean at step t :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\alpha_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Since each reverse step uses the mean from the previous iteration, this bias accumulates throughout the denoising trajectory, ultimately contributing to the overall prediction error.

A.2 POOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

Based on Eq. (7), we take a one-step numerical solver approximates the solution as example. Let $\mathbf{x}^*(\tau)$ denote the ideal trajectory that satisfies this ODE with the true score function $h^*(\mathbf{x}(\tau), \tau)$ and starts from a ground-truth initial condition $\mathbf{x}^*(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}_T)$. Similarly, $\hat{\mathbf{x}}_k$ represent the numerical trajectory computed by a numerical ODE solver at discrete time steps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$ and $\hat{h}$ is the estimated model function. For a specific numerical trajectory from $\hat{\mathbf{x}}_0 - \hat{\mathbf{x}}_N$, we want to calculate its total accumulated trajectory error. Let $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$ be the trajectory error at step k . Consider the transition from τ_k to $\tau_{k+1} = \tau_k - \Delta t$.

Ideal path evolution. Using the variation-of-constants formula for the exact solution over one step (from τ_k down to τ_{k+1}):

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear part $\frac{dz}{d\tau} = f(\tau)z$. Note that the integration limits are backward in time.

Numerical path evolution. A one-step numerical solver approximates the solution. This numerical step deviates from the exact evolution governed by $\hat{h}$ starting from $\hat{\mathbf{x}}_k$. Let $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h})$ be the exact solution at τ_{k+1} of the ODE using $\hat{h}$, starting from $\hat{\mathbf{x}}_k$ at τ_k :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The numerical method introduces a local truncation error δ_k :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) + \delta_k. \quad (\text{A.11})$$

So, we have:

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

Error recurrence. Subtract the ideal evolution Eq. (A.9) from the numerical evolution Eq. (A.12):

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

Accumulated error formulation. We can unroll this recurrence relation from $k = 0$ to $N - 1$. Let $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}), \dots, \mathcal{G}(\tau_{k+1}, \tau_k)$ for $j > k$, and $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$. Applying the recurrence

repeatedly:

$$e_N = \mathcal{G}(\tau_N, \tau_0)e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \cdot \left(\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \quad (\text{A.14})$$

Substituting $\tau_N = 0$, $\tau_0 = T$, $e_N = \hat{\mathbf{x}}_N - \mathbf{x}^*(0)$, and $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$:

$$\hat{\mathbf{x}}_N - \mathbf{x}^*(0) = \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \cdot \left(\underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \quad (\text{A.15})$$

where the propagated initial error term represents the deterministic impact on the final path error of a specific initial error instance, whose statistical origin lies in the distributional Diffused-Prior Gap.

A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

As shown in Eq. (2), the forward process in DDPM (Ho et al., 2020) can be formulated as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

And the reverse process is defined as follows:

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\bar{\alpha}_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon. \quad (\text{A.17})$$

For DDIM (Song et al., 2021), the forward process is identical to that in Eq. (A.16), while its deterministic reverse process (with $\eta = 0$) is defined as follows:

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_\theta(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \left(\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \epsilon_\theta(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Although DDPM and DDIM share the same forward (diffusion) process, DDIM does not simply strip the stochastic term from DDPM's reverse (denoising) process. This is because the two methods employ different coefficient formulations for the predicted noise term.

A.4 DATASET EXAMPLES AND DETAILS

Figure A1 presents representative examples from the three datasets used in this study. As shown in Table A1, each dataset exhibits a balanced distribution across its respective categories. Table A2 further demonstrates that the distribution of images with different object counts is comparable, indicating the absence of object-count bias across datasets. The datasets appear in the same order in both tables, and the category order in Table A1 is defined as follows:

- **ToyShape:** triangle, square, pentagon;
- **SimObject:** mug, apple, clock;
- **RealHand:** finger.

Table A1: Number of objects per category for each dataset.

	ToyShape	SimObject	RealHand
Category 1	20,027	19,916	25,250
Category 2	19,902	20,101	-
Category 3	19,978	19,974	-

Table A2: Number of images by object count(s) per dataset.

	ToyShape	SimObject	RealHand
1 object	10,003	10,022	-
2 objects	10,087	9,965	-
3 objects	9,910	10,013	-
5 objects	-	-	25,250
Total	30,000	30,000	25,250

A.5 EXAMPLES OF COUNTING HALLUCINATIONS

We provide some representative examples of counting hallucinations corresponding to three datasets, as shown in Fig. A2. For the ToyShape and SimObject datasets, the counting hallucinations contain more than two instances of at least one category or do not contain any instances but with normal background, as shown in Fig. A2(a) and Fig. A2(b). In the Realhand dataset, counting-hallucinated samples are characterized by incorrect finger counts (e.g., 4 or 6 fingers), as shown in Fig. A2(c). Notably, the last two images of the first row in Fig. A2(c) appear to display five fingers; however, one finger is partially incomplete and visibly artifacted, effectively resulting in four valid fingers. Consequently, these samples are classified as counting hallucinations.

A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

Figure A3 presents representative examples of YOLO predictions on the RealHand dataset. The YOLO detector accurately identifies fingertips across diverse hand poses and orientations, including both dorsal and palmar views. However, it may still fail to recognize anatomically implausible configurations, such as hands with duplicated thumbs (e.g., the penultimate sub-figure of the first row and the last sub-figure of the last row in Fig. A3(a)) or hands with five middle fingers but without a thumb (e.g., the last sub-figure of the first row and the penultimate sub-figure of the last row in Fig. A3(a)). These abcounting-valid samples are still detected as five-finger hands and therefore remain indistinguishable from anatomically valid ones in terms of finger counts.

Figure A3(b) shows several counting-hallucinated samples where YOLO predicts incorrect fingertip counts (e.g., 3, 4, 6, or 7 fingertips). Figure A3(c) further illustrates several rare edge cases (<1%) in which the detector fails to identify fingertips with unclear boundaries, cracked fingers, or floating artifacts. Such rare cases can be partly mitigated by tuning YOLO’s inference hyperparameters (e.g., IoU and confidence thresholds); however, doing so may compromise detection performance on the majority of counting-valid samples. Since a consistent detection setup is applied across all evaluations on the RealHand dataset, the presence of these rare cases does not affect the validity of our overall conclusions, even though addressing them remains a challenging and low-reward effort.

A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

Figure A4 shows representative examples of three classes of generated hand images: counting-valid, counting-hallucinated, and non-counting failure samples. As illustrated in Fig. 2, the evaluation procedure of the RealHand dataset first applies the counting-ready indicator to separate counting-ready samples (i.e., counting-valid and counting-hallucinated) from images unsuitable for counting. The latter typically exhibit severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers. Counting-ready samples are then further classified as counting-valid or counting-hallucinated based on whether the detected number of valid fingertips equals five.

A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

In the PTI-Hand experiments, we adopt 10 different hand-generation prompts to prompt Stable-Diffusion-3.5-Medium. The prompts are show as follows.

1. A single human hand, front view, all fingers in frame and visible.
2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.
4. Overhead shot of a person’s hand, entire hand in frame, all fingers visible, plain background.
5. One hand, top view, five fingers visible, simple background.
6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

The definition of hallucinations assumes that the model generates patterns that were never observed during training. However, this assumption is difficult to verify for large-scale pretrained diffusion models, where the full training distribution is inaccessible. To make the quantification of counting hallucinations in such cases feasible, we adopt a conservative criterion: we treat any generated hand image containing six or more fingers as a counting-hallucinated sample. This assumption is well justified, while real images may depict fewer than five visible fingers due to occlusion, articulation, or viewpoint, it is highly unlikely for real data to contain hands with six or more fingers. Thus, finger counts ≥ 6 provide a reliable and empirically verifiable signal of counting hallucination. Generated examples can be found in Fig. A5.

A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

In the JDM experiments reported in Table 3, we fine-tune the denoising UNet for 120k steps, compared to 80k steps for the vanilla LDM. This increased training budget is necessary because extending the latent space from three to six channels in JDM introduces a substantial mismatch from the pretrained latent distribution of LDM. As a result, additional optimization is required to close this distributional gap and allow JDM’s modeling capacity to fully materialize.

A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

Table A3: Comparison of failure rates and image quality between finetuning SD-1.5 and finetuning SD-1.5 with joint-diffusion training paradigm on RealHand. The first row (baseline) reports results from finetuning Stable-Diffusion-1.5 on RealHand data. The second row shows results obtained by freezing all intermediate layers of the baseline model and finetuning only the input and output layers with extended channel dimensions. Lower values indicate better performance.

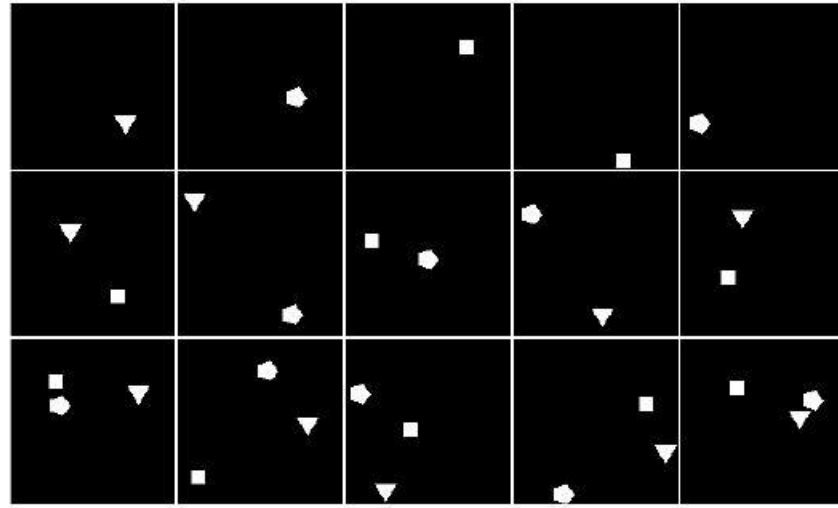
Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
Sampling Steps																		
SD-1.5	0.3362	0.3404	0.3630	0.3396	0.3376	0.3362	0.0059	0.0032	0.0020	0.0050	0.0026	0.0018	61.51	57.71	57.70	58.53	57.57	56.69
SD-1.5 with joint-diffusion finetuning	0.2838	0.3222	0.4145	0.3053	0.3277	0.3558	0.0947	0.0511	0.0487	0.0549	0.0362	0.0327	70.65	79.22	102.39	70.80	72.84	83.07

Importantly, JDM should be viewed as a **novel training paradigm** for improving factual generation in diffusion models rather than as a superior inference strategy. On the one hand, JDM only extends

the input and output channels of the UNet while keeping all sizes of intermediate layers unchanged, resulting in **introduces a negligible inference-time overhead compared to vanilla LDM**. On the other hand, extending a pretrained text-to-image model with additional mask channels **disrupts the manifold geometry of the pretrained latent space and underlines the original denoising distribution**, making direct finetuning such model nontrivial.

Despite this challenge, we still observe promising reductions in counting hallucinations when applying JDM to large-scale pretrained diffusion models under carefully controlled finetuning conditions. In practice, we find that jointly aligning the extended latent space in JDM with the highly rigid latent geometry of large pretrained models is challenging under the limited scale of RealHand. To strike a practical balance, we adopt a two-stage finetuning strategy: we first finetune a Stable-Diffusion-1.5 model on RealHand, then freeze all intermediate UNet layers and finetune only the input and output layers with extended channels. As shown in Table A3, even under this highly constrained finetuning setup, the joint-diffusion paradigm consistently yields lower counting hallucination rates at 25 and 50 sampling steps across both solvers.

Overall, the evidence presented in this work suggests that JDM is a promising direction for enforcing factual consistency in image generation. Lastly, we believe that **exploring its behavior and performance under true large-scale training remains an important and compelling avenue for future research**.



(a) Examples in the ToyShape dataset

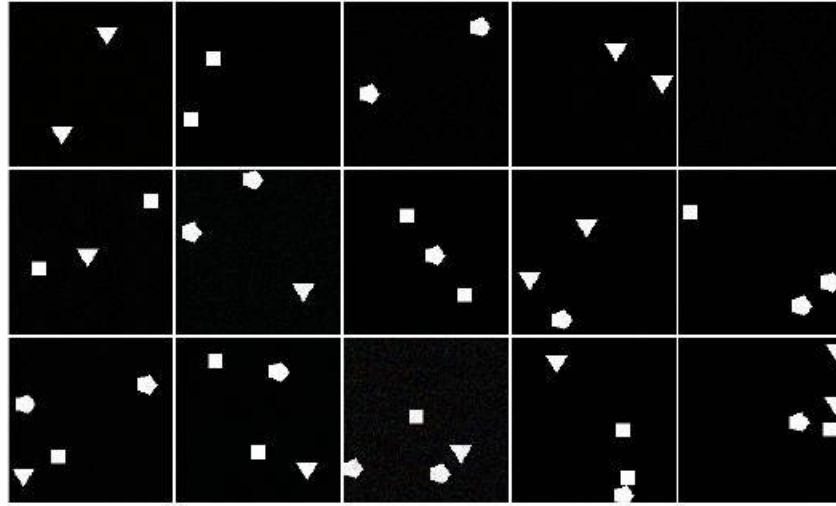


(b) Examples in the SimObject dataset



(c) Examples in the RealHand dataset

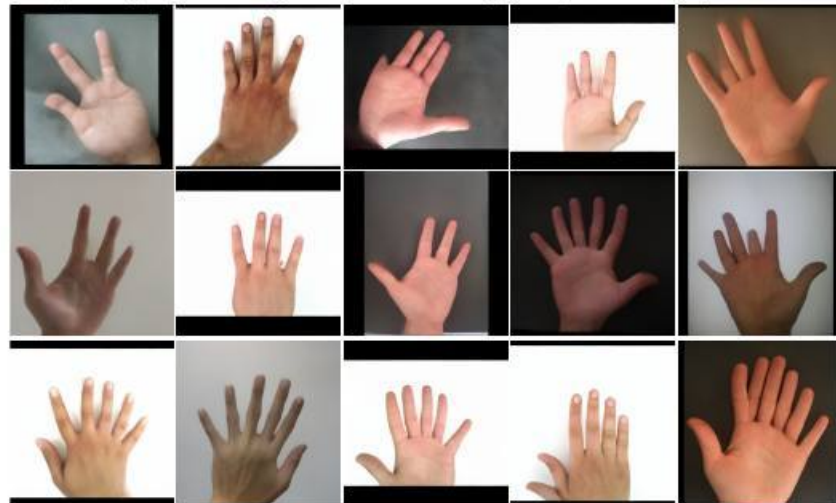
Figure A1: Examples of datasets in the CountHalluSet suite: ToyShape, SimObject, and RealHand.



(a) Examples of counting hallucinations corresponding to the ToyShape dataset

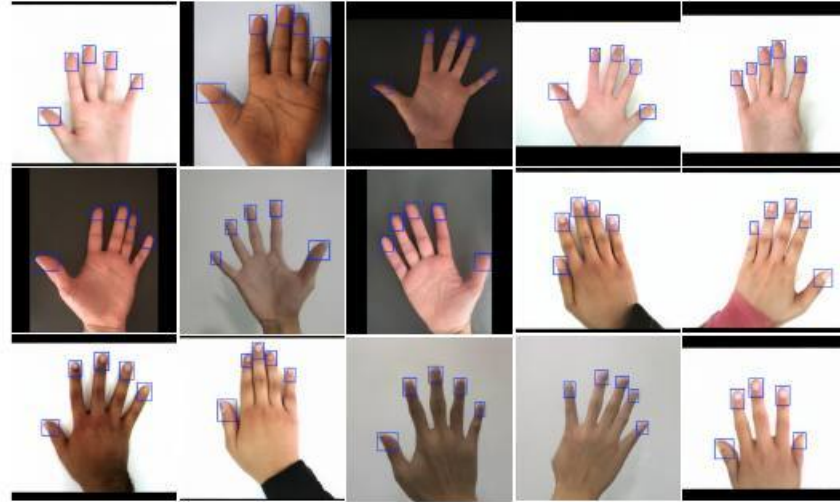


(b) Examples of counting hallucinations corresponding to the SimObject dataset

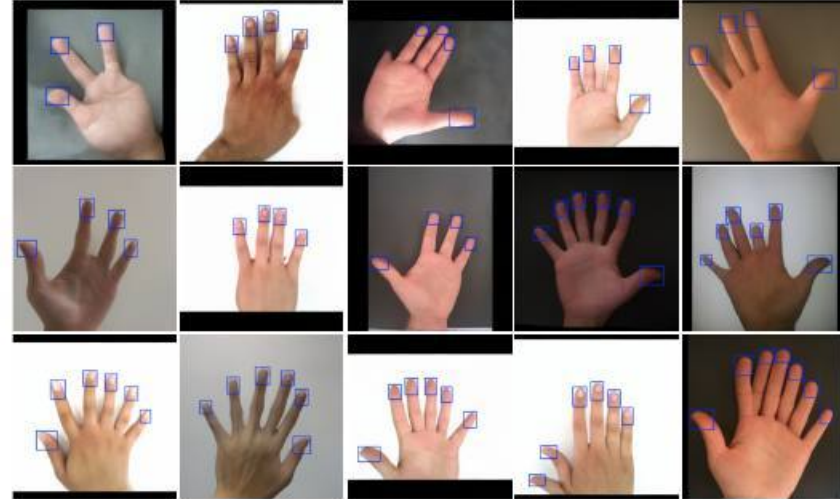


(c) Examples of counting hallucinations corresponding to the RealHand dataset

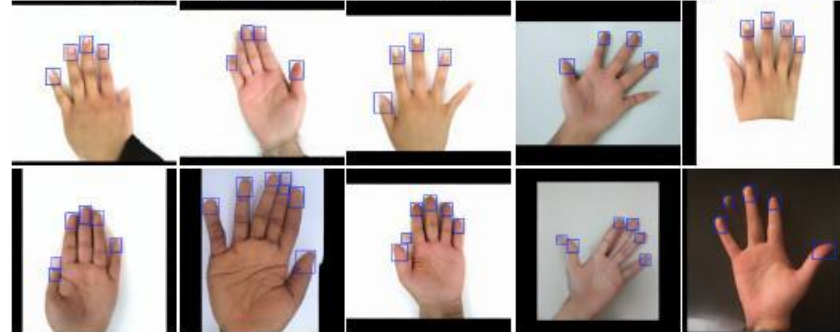
Figure A2: Examples of counting hallucinations corresponding to the ToyShape, SimObject, and RealHand datasets.



(a) Counting-valid YOLO predictions with five fingers

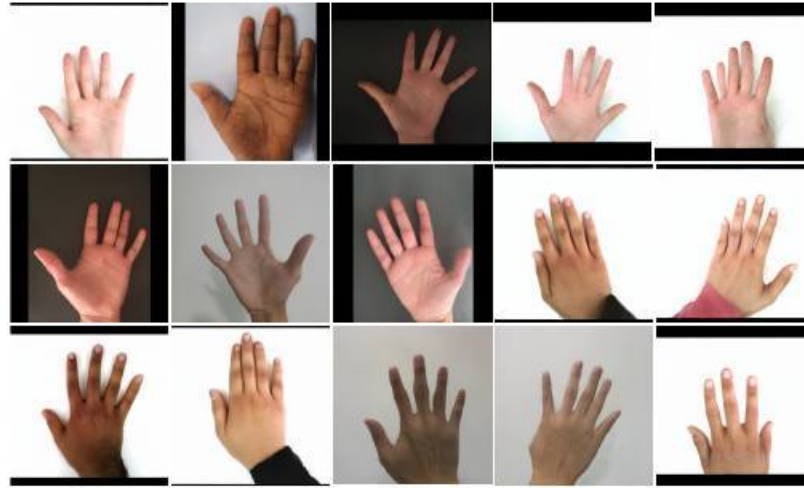


(b) Counting-hallucinated YOLO predictions with not five fingers

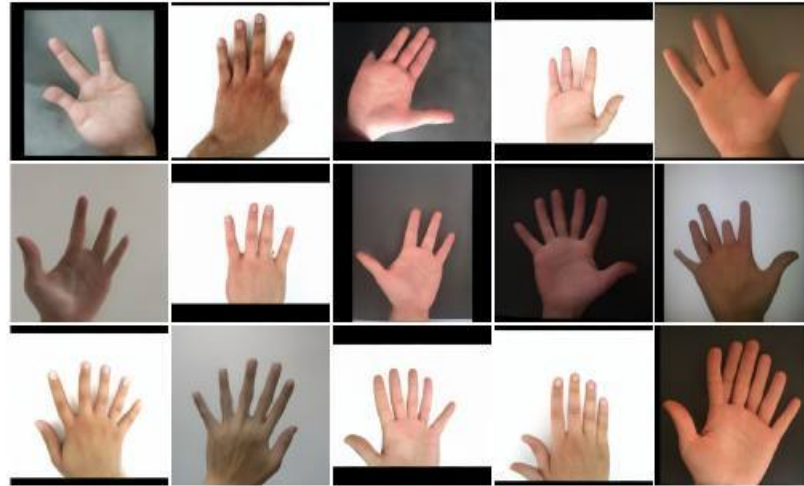


(c) Rare edge cases where YOLO predictions fail

Figure A3: Examples of YOLO predictions on generated RealHand images. (a) Counting-valid samples with five fingertip bounding boxes correctly predicted by YOLO (including cases with two thumbs). (b) Counting-hallucinated samples with incorrect fingertip count predictions (e.g., 3, 4, 6, or 7 detected fingertips). (c) Several rare edge cases where the counting-ready indicator fails to filter out and YOLO struggles to handle, such as blurred, incomplete, or anatomically disconnected fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (not five valid fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A4: Examples of three classes of generated hand images. (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with an incorrect number of valid fingers (e.g., 4 or 6). (c) Non-counting failures: images unsuitable for counting due to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.

1242
1243
1244
1245
1246
1247
1248
1249
1250
1251
1252
1253
1254
1255
1256
1257
1258
1259
1260
1261
1262
1263
1264
1265
1266
1267
1268
1269
1270
1271
1272
1273
1274
1275
1276
1277
1278
1279
1280
1281
1282
1283
1284
1285
1286
1287
1288
1289
1290
1291
1292
1293
1294
1295



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (more than 5 fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A5: Examples of three classes of generated hand images using pretrained Stable-Diffusion-3.5-Medium (PTI-Hand). (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with more than five valid fingers (e.g., 6 or 7). (c) Non-counting failures: images unsuitable for counting due to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.